Cupcast I: A Dixon–Coles Model with Squad-Quality Shrinkage for Forecasting the 2026 FIFA World Cup

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Abstract

I specify a forecasting model for the 2026 FIFA World Cup: a Dixon–Coles goal model [3] fitted by weighted maximum likelihood on a decade of international results, whose team ratings are shrunk toward an empirical prior built from Elo ratings and minutes-weighted, performance-based squad-quality composites. Match probabilities feed a Monte Carlo simulation of the full 48-team tournament under FIFA's official group tiebreakers and Round-of-32 bracket allocation. All tuning decisions (time-decay rate, friendly weighting, shrinkage strength) are made by out-of-sample predictive likelihood and reported here. Validation is the subject of Paper II; the forecast itself of Paper III.

1 Introduction

The design goals are: (i) every match receives a full scoreline distribution, not a point prediction; (ii) every modeling choice is either standard in the literature or selected by out-of-sample evidence; (iii) squad quality enters through measurable on-pitch output, never market valuations; and (iv) the whole pipeline is reproducible from a clean checkout — data fetchers are cache-first, every stochastic step is explicitly seeded, and the published numbers regenerate from one command.

The pipeline is: match history → Elo ratings (§3) → Dixon–Coles fit (§4) → shrinkage toward a squad-informed prior (§6) → 50,000 tournament simulations (§7).

2 Data

2.1 International match history

The training table is built from the community-maintained CC0 dataset of international results [7], current through June 9, 2026, totalling 49,400 matches since 1872 with venues and neutral-site flags. Cross-checks and competition metadata (rounds, fixtures with team identifiers) come from API-Football [1] for the 2022–2024 seasons. Non-FIFA internationals are removed by restricting to teams that appear in World Cup qualification or a Nations League within the window; the Dixon–Coles training set since January 2015 holds 10,577 matches across 226 teams.

2.2 Squads and player data

Confirmed 26-man squads for all 48 teams (registration deadline June 2, 2026) are parsed from the structured player templates of the consolidated squads page [14]: 1,247 players with shirt number, position, age at kickoff, caps, and club. (Argentina registered 25 players; the source is faithful on this point.) Player-level performance over the 2025–26 club season comes from the npxG family of metrics across the big five European leagues: FBref's Opta-sourced tables (npxG+xAG, plus goalkeeper post-shot expected goals) [10] where available, with Understat (npxG+xA) [12] as an equivalent source — both are standardised within their own player pool, so the composite scale is invariant to which one feeds it. Coverage tiers and the fallback for squads based in uncovered leagues are described in §5. No market-value data is used anywhere.

2.3 Bookmaker odds and club data

Outright championship odds across bookmakers are snapshotted from The Odds API [11], with proportional overround removal; the model–market comparison appears in Paper III. Club league results with Pinnacle closing odds [5] support the engine-level validation tier in Paper II.

3 Elo Ratings

The rating input follows the World Football Elo system [15], whose predictive value for match outcomes is established empirically by Hvattum and Arntzen [6]. After each match the rating of team i updates as

$$R_i' = R_i + K\,G\,(W - W_e), \qquad W_e = \frac{1}{1 + 10^{-(R_i - R_j + H)/400}}, \tag{1}$$

where W is the realised result, $H{=}100$ the home adjustment (zero at neutral venues), G the goal-difference multiplier (1, 1.5, then $1.75 + (n-3)/8$ for margins $n \geq 3$), and K encodes importance: 60 for World Cup finals matches, 50 for continental finals tournaments, 40 for World Cup and continental qualifiers and the Nations Leagues, 30 for other tournaments, 20 for friendlies. Ratings are computed from the raw match history rather than scraped, which yields reproducible as-of-date ratings for backtesting; the June 2026 ladder (Spain 2219, Argentina 2190, France 2125, England 2086, Brazil 2069, ...) matches the published ordering.

4 The Goal Model

Goals scored by teams i and j are conditionally Poisson with rates

$$\lambda_i = \exp\bigl(\mu + \alpha_i - \beta_j + \gamma\,h_i\bigr), \qquad \lambda_j = \exp\bigl(\mu + \alpha_j - \beta_i + \gamma\,h_j\bigr), \tag{2}$$

where α and β are attack and defense ratings and h is a host indicator: in training it marks genuine home fixtures; at the 2026 World Cup it is nonzero only for Mexico, the United States, and Canada playing inside their own country (the schedule keeps all three at home for the entire group stage). Low-score dependence is corrected by the Dixon–Coles adjustment [3]

$$\mathbb{P}(X{=}x, Y{=}y) = \tau_\rho(x,y)\,\frac{e^{-\lambda_i}\lambda_i^x}{x!}\,\frac{e^{-\lambda_j}\lambda_j^y}{y!}, \tag{3}$$

with τ_ρ reweighting the $\{0,1\}^2$ scorelines, addressing the draw deficit of the independent-Poisson model of Maher [9].

4.1 Estimation

Parameters are estimated by weighted maximum likelihood with analytic gradients under L-BFGS-B; identifiability is imposed through a quadratic penalty pinning the means of α and β at zero, and a weak ridge (10^{-2}) keeps teams with negligible decayed sample bounded. The 483-parameter fit on 10,577 matches converges in roughly one second. Fitted globals as of June 11, 2026: $\mu = 0.062$, home advantage $\gamma = 0.242$ (a +27% goal rate), $\rho = -0.046$.

4.2 Recency weighting

Matches are weighted by $\phi(t) = \exp(-\xi\,\Delta t)$. The decay rate and a candidate friendly down-weight were tuned jointly by out-of-sample log-loss over eight half-year rolling folds (January 2022 – June 2026, $n = 4167$ matches):

Half-life	friendly ×0.3	friendly ×0.6	friendly ×1.0
1.5y	0.9246	0.8962	0.8737
2.5y	0.8646	0.8631	**0.8623**
3.5y	0.8854	0.8633	0.8626
5.0y	0.8837	0.8643	0.8637

The optimum is flat between half-lives of 2.5 and 3.5 years, consistent with Ley et al. [8]; 2.5 years is used. Notably, down-weighting friendlies *strictly worsens* predictive likelihood at every half-life. The Elo K-hierarchy intuition — friendlies matter less — concerns rating *stakes*, not informational content: friendly scorelines are evidently as informative about team strength as competitive ones, so they enter the likelihood at full weight.

5 Squad-Quality Composite

Each player's quality is the league-pooled z-score of non-penalty expected goal involvement per 90 minutes (npxG+xAG from [10], or npxG+xA from [12]) over the 2025–26 club season (minimum 450 minutes). Squad composites aggregate player quality using *expected tournament minutes* as weights. Expected minutes follow a depth-chart heuristic on a 4-3-3 baseline: within each position group, starters are the most-capped players; goalkeepers play the full match, outfield starters retain 85% of their slot's minutes, and the bench splits the remainder in proportion to caps. Squad-level age is also computed minutes-weighted. Players not matched to the covered leagues are imputed at the matched pool's 20th percentile; a team's *coverage* is the share of its expected minutes that matched. Teams under 35% coverage fall back to the Elo-only prior tier below, and per-team coverage is reported with the forecast. The caps-based depth chart is a deliberate v1 simplification — it under-ranks recently emerged starters — and is the first refinement candidate listed in Paper III's limitations.

6 Shrinkage Toward an Empirical Prior

Fitted ratings are blended with a cross-team empirical prior,

$$\tilde{\alpha}_i = w_i\,\hat{\alpha}_i + (1 - w_i)\,\alpha_i^{\text{prior}}, \qquad w_i = \frac{n_i^{\text{eff}}}{n_i^{\text{eff}} + k}, \tag{4}$$

where n_i^{eff} is the decay-weighted match count. The prior is the prediction of a weighted least-squares regression of fitted ratings on standardised Elo — plus the squad composite where coverage permits — so that well-identified teams define the relationship and thinly observed teams are pulled toward it: the partial-pooling logic of Baio and Blangiardo [2] in a transparent two-stage form. The strength k was tuned on the same rolling folds as §4.2:

k	0	**10**	25	50	100
Log-loss	0.8623	**0.8615**	0.8641	0.8671	0.8706

Light shrinkage (k=10) helps; anything stronger is worse than none. Under the tuned decay an international regular accumulates $n^{\text{eff}} \approx 40$, so established teams keep $\geq 80\%$ of their fitted rating while a thin-history qualifier leans on the prior.

7 Tournament Simulation

Each of the 50,000 simulations plays all 72 group matches by sampling scorelines from the Dixon–Coles grid, ranks groups under the FIFA tiebreaker chain (points, goal difference, goals for, then head-to-head points/GD/GF among the tied teams, then drawing of lots — fair-play points are not modelled, a documented simplification), ranks the twelve third-placed teams by the same criteria to find the eight qualifiers, and maps qualifiers into the official Round-of-32 bracket [13].

Third-place allocation. FIFA's Annex C [4] fixes the assignment of qualified thirds to the eight bracket slots for each of the 495 possible combinations. The published table is not machine-readable; the simulator instead solves each combination by deterministic constraint matching (most-constrained slot first, alphabetical preference). This reproduces FIFA's published worked example exactly, and a unit test verifies that all 495 combinations yield valid assignments. Any divergence from unpublished rows of Annex C would only permute which group winner faces which third — a second-order effect on team-level probabilities.

Knockouts. Ties after regulation continue into extra time, modelled as a Dixon–Coles grid at one-third rates, then a penalty shootout

$$p = \tfrac{1}{2} + \operatorname{clip}\big(0.02\,(z_a - z_b),\ \pm 0.05\big), \tag{5}$$

where z is the first-choice goalkeeper's shot-stopping over-performance (PSxG± per 90, standardised across tournament keepers [10]); without keeper data the shootout is an even coin. Host advantage applies in knockouts only when a host plays inside its own country, which the bracket fixes per match.

8 Reproducibility

Every fetcher caches raw responses on first call and the pipeline reads only from cache; API-Football calls pass through a persisted request ledger. All random draws flow from explicitly passed seeds (NumPy `Generator`); the full pipeline — fit, shrinkage, 50,000 simulations, reports — runs in well under a minute on commodity hardware via `python -m cupcast.report.build`, and the tuning grids in §4.2 and §6 regenerate via the documented validation modules.

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